Numerical Analysis

Team_7

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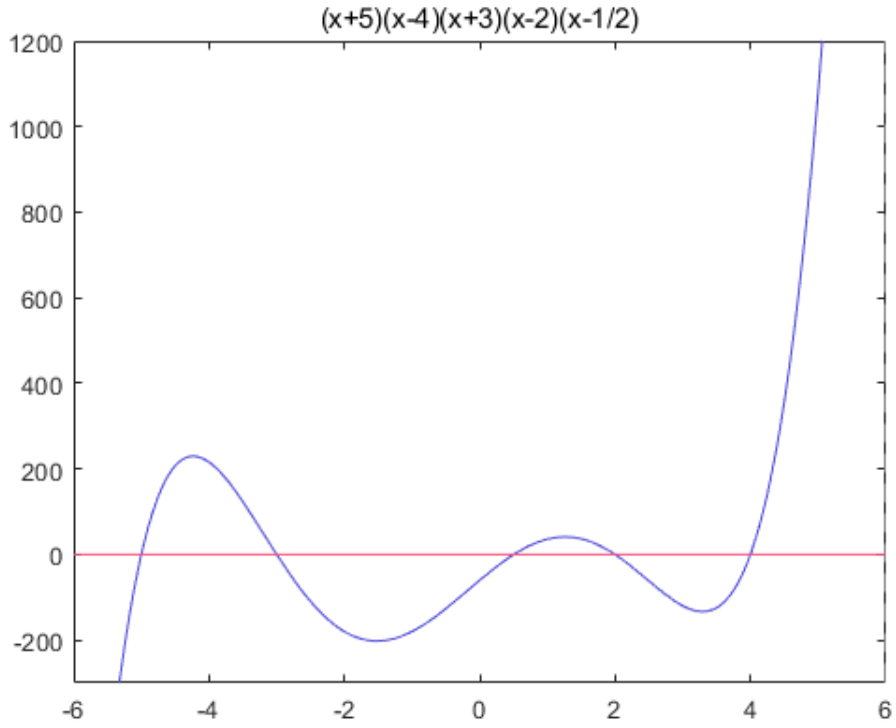
Question 1

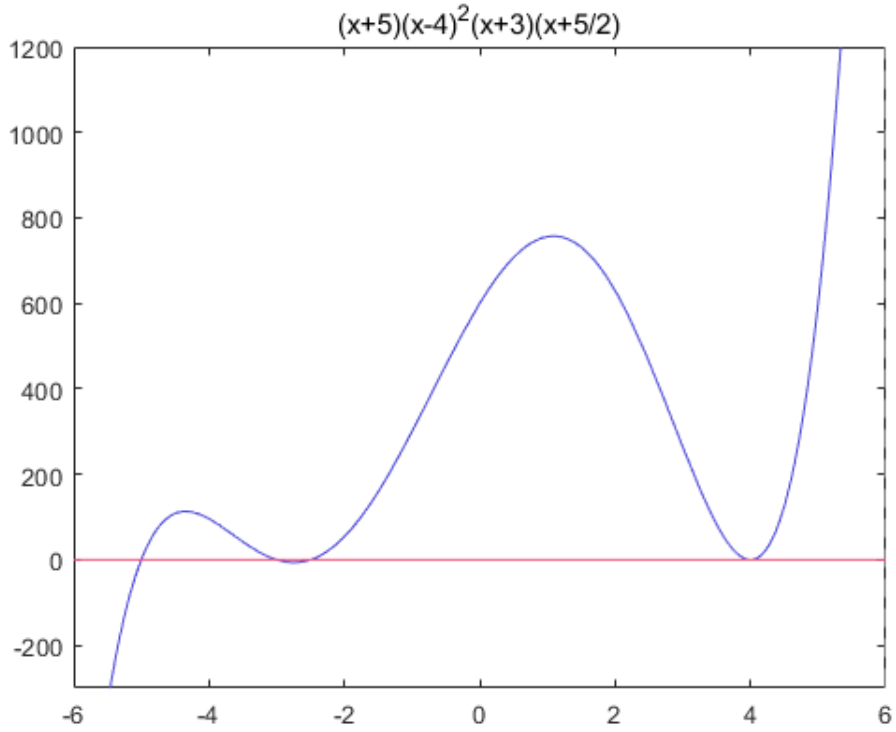
1.1

Design two nonlinear algebraic equations (degree at least 5 , with no less than 3 zeros) and find all their zeros (error less than 0.001).
So,we firstly design the quintic algebric equations as follows.

$$y_1 = (x+5)(x-4)(x+3)(x-2)(x-\frac{1}{2})$$
$$y_2 = (x+5)(x-4)^2(x+3)(x+\frac{5}{2})$$

we use the Matlab to print the figure of the equation.





1.2

By analyzing the y_1 images, we can easily the minimum distance of zero is less than $\frac{1}{2}$, and the minimum distance of zero in y_2 is less than $\frac{1}{2}$ too.Thus, We divide the whole interval$[-6,6]$ into a number of cells with a length of 1/2,and using the method of bisection solve for zero separately.

```
x0 = zeros(1,5);
k = 0;
```

```
L = 1/2;%分割长度
x0 = unique(x0);
 for i = -6:L:6
      if(y(i)*y(i+L)<=0)
            k = k + 1;
            x0(k) = f_Bis(y,i,i+L,0.001);%k为第k的零点
      end
 end
fprintf("零点的横坐标为:")

%二分法具体函数
function z=f_Bis(f,a,b,accurate)
p=(a+b)/2;
n=0;
z=0;
while (abs(f(p)-0)>accurate)
      if(f(p)==0)%判断端点值是否为零点
            z=p;
            break;
      end
      if(f(a)==0)
            z=a;
            break;
      end
      if(f(b)==0)
            z=b;
            break;
      end
      if(f(a)*f(p)<0)
            b=p;
            p=(a+b)/2;
      else
            a=p;
            p=(a+b)/2;
      end
      z=p;
      n=n+1;
end
n;
end
```

Because of the existence of multiple zeros,Therefore, when solving the zeros, we should also consider the order of the zeros.

We take the derivative of the function, we get the first derivative of the function, and then we determine whether the zero at the first derivative is zero, and if it's zero, then it's the second zero, and so on.

函数的一阶导数值：

dy_x =

$$\left(405 \quad -49 \quad \frac{845}{16} \quad 0\right)$$

函数的二阶导数值：

ddy_x = (−909 175 221 819)

zeros	-5.0000	-3.0000	-2.5000	4.0000
order	1.0000	1.0000	1.0000	2.0000

```
function B = Judge(y,x0)
[C,R] = size(x0);
k = 1;
syms x
y1 = (x+5)*(x-4)*(x+3)*(x-2)*(x-1/2);
y=(x+5)*(x-4)^2*(x+3)*(x+5/2);%该函数只能键入，用参数输入时无法识别
f = y;
dy = diff(f)%求一阶导
ddy = diff(f,2);求二阶导
fprintf("函数的一阶导数值:")
dy_x = subs(dy,x,x0)
dy_x =double(dy_x);%强制转换
fprintf("函数的二阶导数值:")
ddy_x = subs(ddy,x,x0)
ddy_x = double(ddy_x);
for i=1:R
      if(dy_x(i) == 0)
            x0(2,k) = 2;
            if(ddy_x(i) == 0)
                  x0(2,k) = 3;
            end
      else%此处只判断到三阶零点
      x0(2,k) = 1;
      end
      k = k + 1;
end
B = x0;
```

```
end
```

Question 2

Find the numerical solution of the following nonlinear equations using iteration method and Newton method respectively and show your analysis of error by using norms of vector and matrix and demonstrate your computer code

$$\begin{cases} x_1^2 - 10x_1 + x_2^2 + 8 = 0 \\ x_1 x_2^2 + x_1 - 12x_2 + 7 = 0 \end{cases}$$

2.1

Fixed-point iteration method:

We have the following system of equations $\begin{cases} y_1 = f_1(x1, x2) \\ y_2 = f_2(x1, x2) \end{cases}$,we set the initial point $P_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$,the iterative formula is $\begin{matrix} x_1^{(n+1)} = f_1(x_1^{(n)}, x_2^{(n)}) \\ x_2^{(n+1)} = f_2(x_1^{(n)}, x_2^{(n)}) \end{matrix}$.

In order to compare with the Newton-Raphson iteration method,The error is

$$\left\| \begin{bmatrix} x_1^{(n+1)} \\ x_2^{(n+1)} \end{bmatrix} - \begin{bmatrix} x_1^{(n)} \\ x_2^{(n)} \end{bmatrix} \right\| = \left\| \begin{bmatrix} f_1(x_1^n, x_2^n) - f_1(x_1^{(n-1)}, x_2^{(n-1)}) \\ f_2(x_1^n, x_2^n) - f_2(x_1^{(n-1)}, x_2^{(n-1)}) \end{bmatrix} \right\|$$

```
N = 100;% 设置最大迭代次数
epsilon = 1e-9;% 设置迭代的精度
%Fixed-point iteration
x(1,:) = [1;1];%迭代初始点
for i = 1:N
      x(i+1,1) = 1/10*(x(i,1)^2+x(i,2)^2+8);
      x(i+1,2) = 1/12*(x(i,1)+x(i,1)*x(i,2)^2+7);
      dx_Fp(i,:) = x(i+1,:) - x(i,:);
      if(sum(dx_Fp(i,:).^2)< epsilon)
            break;
      end
end
x_Fp = x
```

Newton-Raphson Iteration

Firstly,we set the initial point $x_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$, the $x^{(k+1)} - x^{(k)} = \frac{-F(x^{(k)})}{J(x^{(k)})}$.

we set the $d^{(k)} = x^{(k+1)} - x^{(k)}$.

Secondly,The termination condition of the iteration is $\left\| d^{(k)} \right\| < \varepsilon$

```
%Newton-Raphson iteration
syms x1 x2;
f1(x1,x2) = x1^2 - 10*x1 + x2^2 + 8;
f2(x1,x2) = x1*x2^2 + x1 - 12*x2 + 7;

F(x1,x2) = [f1;f2];

x_solve(:,1) = [1;1];%迭代初始点

J(x1,x2) = jacobian(F,[x1 x2]);

dx_Nr =[0;0];          % X(n+1)-X(n) = d(X); 第一行表示第一次迭代X(x1,x2)值与前一值的插值
for i = 1:N                    % J*d(x) = -F(X) => d(x) = J \ -F(x)
      dx_Nr(:,i) = J(x_solve(1,i),x_solve(2,i))\(-F(x_solve(1,i),x_solve(2,i)));
      if(sqrt(sum(dx_Nr(:,i).^2)) < epsilon) %
            x_solve(:,i+1) = x_solve(:,i) + dx_Nr(:,i);
            break;
      else
            x_solve(:,i+1) = x_solve(:,i) + dx_Nr(:,i);
      end
end
dx_Nr = dx_Nr';
x_Nr = x_solve'
```

Result:

```
      x_Fp = 10×2
      1.0000      1.0000      %误差
      1.0000      0.7500      0.0625
      0.9563      0.7135      0.0032
      0.9424      0.7036      2.9203e-04
      0.9383      0.7007      2.4534e-05
      0.9371      0.6999      2.0193e-06
      0.9368      0.6997      1.6524e-07
```

```
    0.9367    0.6996    1.3499e-08
    0.9367    0.6996    1.1022e-09
    0.9367    0.6996    8.9990e-11
    %迭代 9次

x_Nr = 6×2

    1.0000    1.0000
    0.9211    0.6842    0.3255
    0.9366    0.6995    0.0218
    0.9367    0.6996    9.2006e-05
    0.9367    0.6996    1.6056e-09
    0.9367    0.6996    5.7481e-17
    %迭代 5次
% it is easy to find that the Newton method is better than fix-point
method
```

We generalize the two-dimensional model to the three-dimensional model,

$$\begin{cases} 5x_1 - x_2^2 + x_1x_3 = 2 \\ x_1x_3 + 4x_2 - x_3 = 0 \\ x_1^2 + x_2 - 3x_3 = 1 \end{cases}$$

we only need to expand a variable，The changed details are as follows:

```
%Fixed-point iteration
x(1,1) = 1;x(1,2) = 1;x(1,3) =1;%迭代初始点
for i = 1:N
    x(i+1,1) = 1/5*(x(i,2)^2-x(i,1)*x(i,3)+2);
    x(i+1,2) = 1/4*(x(i,3)-x(i,1)*x(i,3));
    x(i+1,3) = 1/3*(x(i,1)^2+x(i,2)-1);
    dx_Fp(i,:) = x(i+1,:) - x(i,:);
    if(sum(dx_Fp(i,:).^2)< epsilon)
        break;
    end
end
x_Fp = x


%Newton-Raphson iteration
syms x1 x2 x3;
f1(x1,x2,x3) = 5*x1-x2^2+x1*x3-2;
f2(x1,x2,x3) = x1*x3+4*x2-x3;
f3(x1,x2,x3) = x1^2+x2-3*x3-1;

```

```
F(x1,x2,x3) = [f1;f2;f3];

x_solve(:,1) = [0;0;0];%迭代初始点

J(x1,x2,x3) = jacobian(F,[x1 x2 x3]);

dx_Nr =[1;1;1];          % X(n+1)-X(n) = d(X); 第一行表示第一次迭代X(x1,x2)值与前一值的差值
for i = 1:N                    % J*d(x) = -F(X) => d(x) = J \ -F(x)
      dx_Nr(:,i) = J(x_solve(1,i),x_solve(2,i),x_solve(3,i))\(-F(x_solve(1,i),x_solve(2,i),x_solve(3,i)));
      if(sqrt(sum(dx_Nr(:,i).^2)) < epsilon)
            x_solve(:,i+1) = x_solve(:,i) + dx_Nr(:,i);
            break;
      else
            x_solve(:,i+1) = x_solve(:,i) + dx_Nr(:,i);
      end
end
dx_Nr = dx_Nr';
x_Nr = x_solve'
```

Question 3

3.1

(1)Generate matrix $A(50 \times 50)$ and vector $b(50 \times 1)$ by using random number, and solve the linear system $AX = b$ by using Gaussian partial pivoting elimination algorithm.

3.2

There are two important points:\
1)Generate A nonsingular matrix to make it solvable;\
2)Using sort algorithm to find the partial pivot.

```
n=50;
while(1)
A=round(rand(n)*99)+1;    %生成n阶方阵，元素取值范围为1-100
%判断矩阵是否奇异
if d=det(A)
```

```
        break
end
end
b=round(rand(n,1)*99)+1;    %生成n*1向量，元素取值范围为1-100
X=[n,1];
T=[A,b];
Temp=[n+1,1];     %用作交换两行

for i=1:n
        for j=i+1:n    %选取绝对值最大的为列主元
                if(abs(T(j,i))>abs(T(i,i)))
                        max=T(j,:);
                        Temp=T(i,:);
                        T(i,:)=T(j,:);
                        T(j,:)=Temp;
                end
        end

        for k=i+1:n    %计算得到上三角矩阵
        m_ik=T(k,i)/T(i,i); %计算系数
        T(k,i+1:n+1)=T(k,i+1:n+1)-(T(i,i+1:n+1).*m_ik);
        T(k,i)=0;
        end
end

X(n,1)=T(n,n+1)/T(n,n);    %计算x[n,1]
for i=2:n    %计算x[1:n-1,1]
        X(n-i+1,1)=(T(n-i+1,n+1)-T(n-i+2,n-i+2:n)*X(n-i+2:n,1))/T(i,i);
end
```

Randomly generate tridiagonal matrix (30 * 30), and use Jacobi iteration and Gauss Seidel iteration to solve them respectively. The error of the solution in terms of norm is required to be less than 1/1000.

The most important premise to solve the problem is we can get a matrix which match the Jacobi iterative and Gauss-Seidel iterative conditions.

A strictly diagonally dominant is eligibility.

```
 %创建30阶随机方阵，元素取值1-100
A=round(rand(30)*99)+1;
D=zeros(30);
U=zeros(30);
```

```
L=zeros(30);
B=round(rand(30,1)*99)+1;

%将D-U-L变为严格对角占优矩阵，使得满足迭代条件
for j=1:29
        U(j,j+1)=-A(j,j+1);
        L(j+1,j)=-A(j+1,j);
end
D(1,1)=sum(abs(A(1,1:2)))+10*rand(1);
for i=2:29
        D(i,i)=sum(abs(A(i,i-1:i+1)))+10*rand(1);
end
D(30,30)=sum(abs(A(30,29:30)))+10*rand(1);

T=((D-L)^(-1))*U;
C=(D-L)^(-1);

e=max(abs(eig(T)));    %检查系数矩阵特征值

Xn=zeros(30,1);    %存储当前的迭代结果
Xn_1=ones(30,1);    %存储上一次的迭代结果
while sqrt(sum(abs(Xn-Xn_1).^2))>0.001
        Xn_1=Xn;
        for j=1:30
        Xn(j,1)=T(j,:)*Xn+C(j,:)*B;
        end
        fprintf('Xn = %f\n',Xn);
end
```

```
%创建30阶随机方阵，元素取值1-100
A=round(rand(30)*99)+1;
D=zeros(30);
U=zeros(30);
L=zeros(30);
B=round(rand(30,1)*99)+1;

%将D-U-L变为对角占优矩阵，使得满足迭代条件
for j=1:29
        U(j,j+1)=-A(j,j+1);
        L(j+1,j)=-A(j+1,j);
end
D(1,1)=sum(abs(A(1,1:2)))+10*rand(1);
for i=2:29
```

```
        D(i,i)=sum(abs(A(i,i-1:i+1)))+10*rand(1);
end
D(30,30)=sum(abs(A(30,29:30)))+10*rand(1);

T=(D^-1)*(L+U);
e=max(abs(eig(T)));    %检查系数矩阵特征值

Xn=zeros(30,1);    %存储当前的迭代结果
Xn_1=ones(30,1);    %存储上一次的迭代结果
while sqrt(sum(abs(Xn-Xn_1).^2))>0.001
        Xn_1=Xn;
        Xn=(D^-1)*(L+U)*Xn_1+(D^-1)*B;
        fprintf('Xn = %f\n',Xn);
end
```

Question 5

Use appropriate polynomial interpolation to approximate the following ellipse.

(1)$x^2+y^2=5^2$;

(2)$\rho(\theta)=2(1-sin\theta)$(心形线).

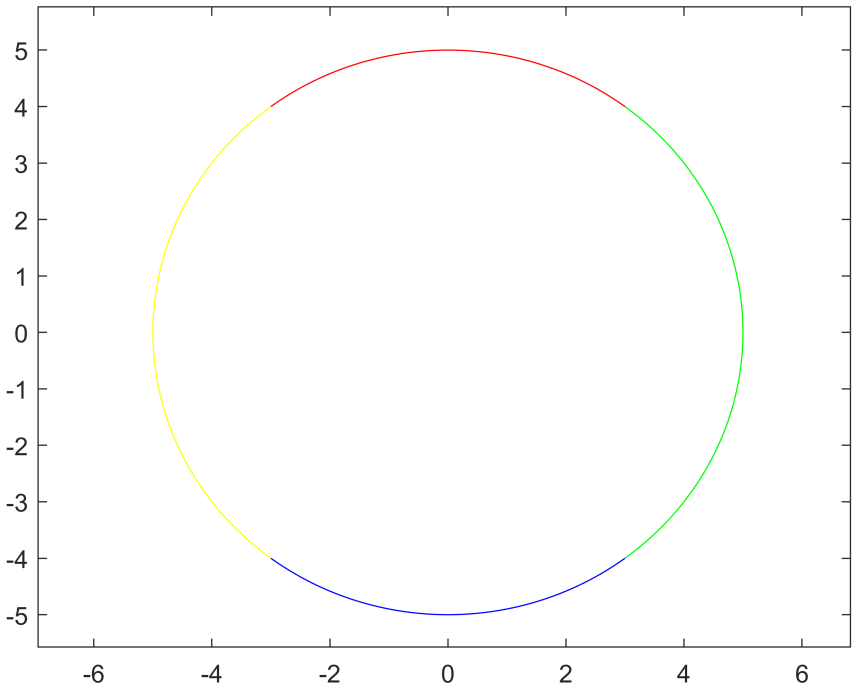
It's useful that using Spline Interpolation for the curve which is differentiable.But

(1)There are some point whose derivative are infinity for $f(x)$.We can divide the cuvre into several parts which includes one special point at least.And exchanging independent variable and dependent variable make some special points differentiable.

(2)Some points are nondifferentiable but continuous,thay can be the endpoints.

```
%对上下两段插值
x_i=-3:0.5:3;
y1=(25-x_i.^2).^(1/2);
y2=-(25-x_i.^2).^(1/2);
xx_i=-3:0.1:3;
yy1=spline(x_i,y1,xx_i);
yy2=spline(x_i,y2,xx_i);
%对左右两段插值
y_i=-4:0.5:4;
x1=(25-y_i.^2).^(1/2);
x2=-(25-y_i.^2).^(1/2);
yy_i=-4:0.1:4;
xx1=spline(y_i,x1,yy_i);
```

```
xx2=spline(y_i,x2,yy_i);
%画出图像
plot(xx_i,yy1,xx_i,yy2,xx1,yy_i,xx2,yy_i);
```




```
theta_1=0:pi/36:pi/2;
x=rho(theta_1).*cos(theta_1);
y=rho(theta_1).*sin(theta_1);
theta_11=0:pi/108:pi/2;
xx_1=rho(theta_11).*cos(theta_11);
yy_1=spline(x,y,xx_1);

theta_2=pi/2:pi/36:pi;
x=rho(theta_2).*cos(theta_2);
y=rho(theta_2).*sin(theta_2);
theta_22=pi/2:pi/108:pi;
xx_2=rho(theta_22).*cos(theta_22);
yy_2=spline(x,y,xx_2);

theta_3=pi:pi/36:(8*pi)/6;
x=rho(theta_3).*cos(theta_3);
y=rho(theta_3).*sin(theta_3);
theta_33=pi:pi/108:(8*pi)/6;
yy_3=rho(theta_33).*sin(theta_33);
xx_3=spline(y,x,yy_3);

```

```
theta_4=(8*pi)/6:pi/36:(10*pi)/6;
x=rho(theta_4).*cos(theta_4);
y=rho(theta_4).*sin(theta_4);
theta_44=(8*pi)/6:pi/108:(10*pi)/6;
xx_4=rho(theta_44).*cos(theta_44);
yy_4=spline(x,y,xx_4);

theta_5=(10*pi)/6:pi/36:2*pi;
x=rho(theta_5).*cos(theta_5);
y=rho(theta_5).*sin(theta_5);
theta_55=(10*pi)/6:pi/108:2*pi;
yy_5=rho(theta_55).*sin(theta_55);
xx_5=spline(y,x,yy_5);

plot(xx_1,yy_1,xx_2,yy_2,xx_3,yy_3,xx_4,yy_4,xx_5,yy_5);
function rho=rho(theta)
rho=2.*(1-sin(theta));
end
```

Question 5